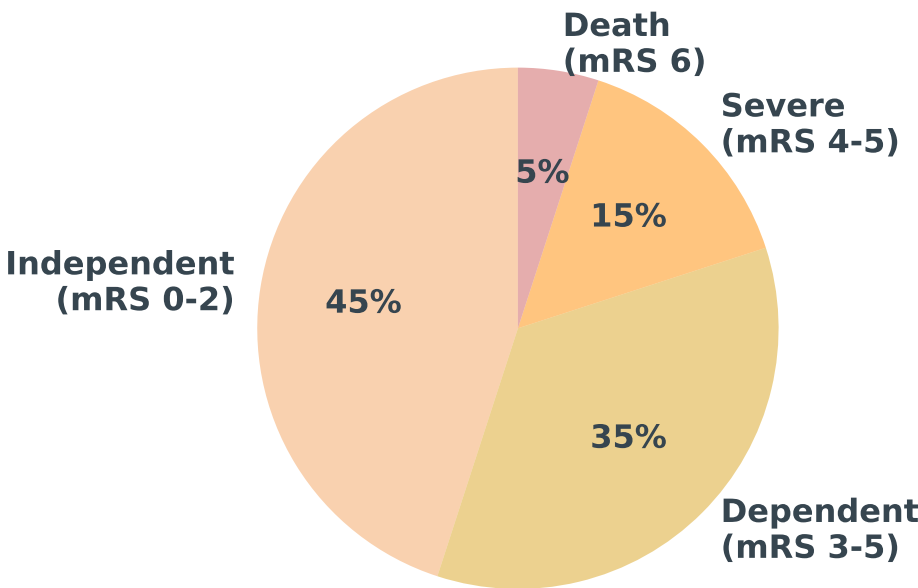


Clinical Problem & Motivation

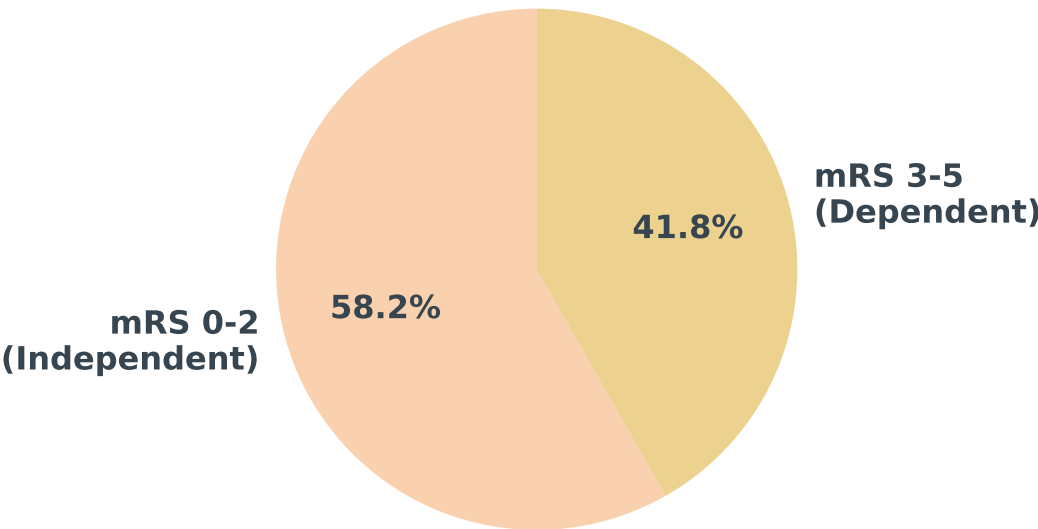
Current Prediction Methods Accuracy

Global Stroke Outcome Distribution



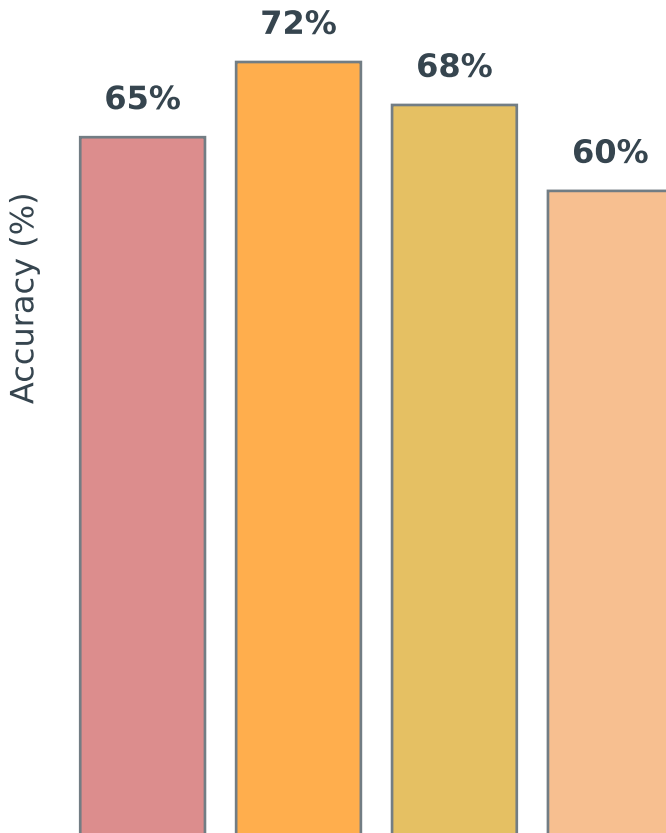
- 15 million people annually experience stroke worldwide
- Leading cause of long-term disability

Our Outcome Distribution



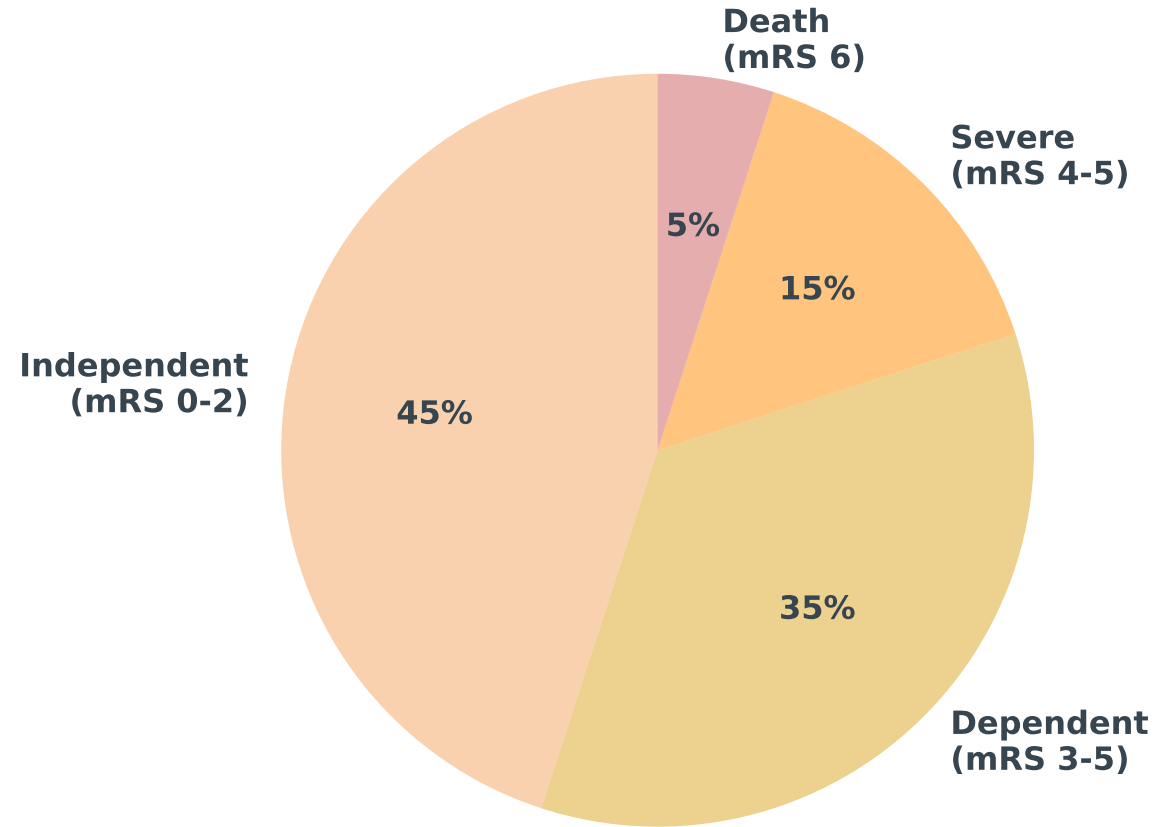
- Guides treatment plans, rehabilitation, and family counseling
- The Modified Rankin Scale (mRS): Measures post-stroke functional independence

mRS 0-2: Independent; 3-5: Dependent



- Comprehensive analysis using 455 clinical MRI scans from 91 patients
 - Real-world data with documented mRS outcomes
 - Direct clinical relevance, immediate impact on patient care

Stroke: A Major Health Burden - Detailed Analysis



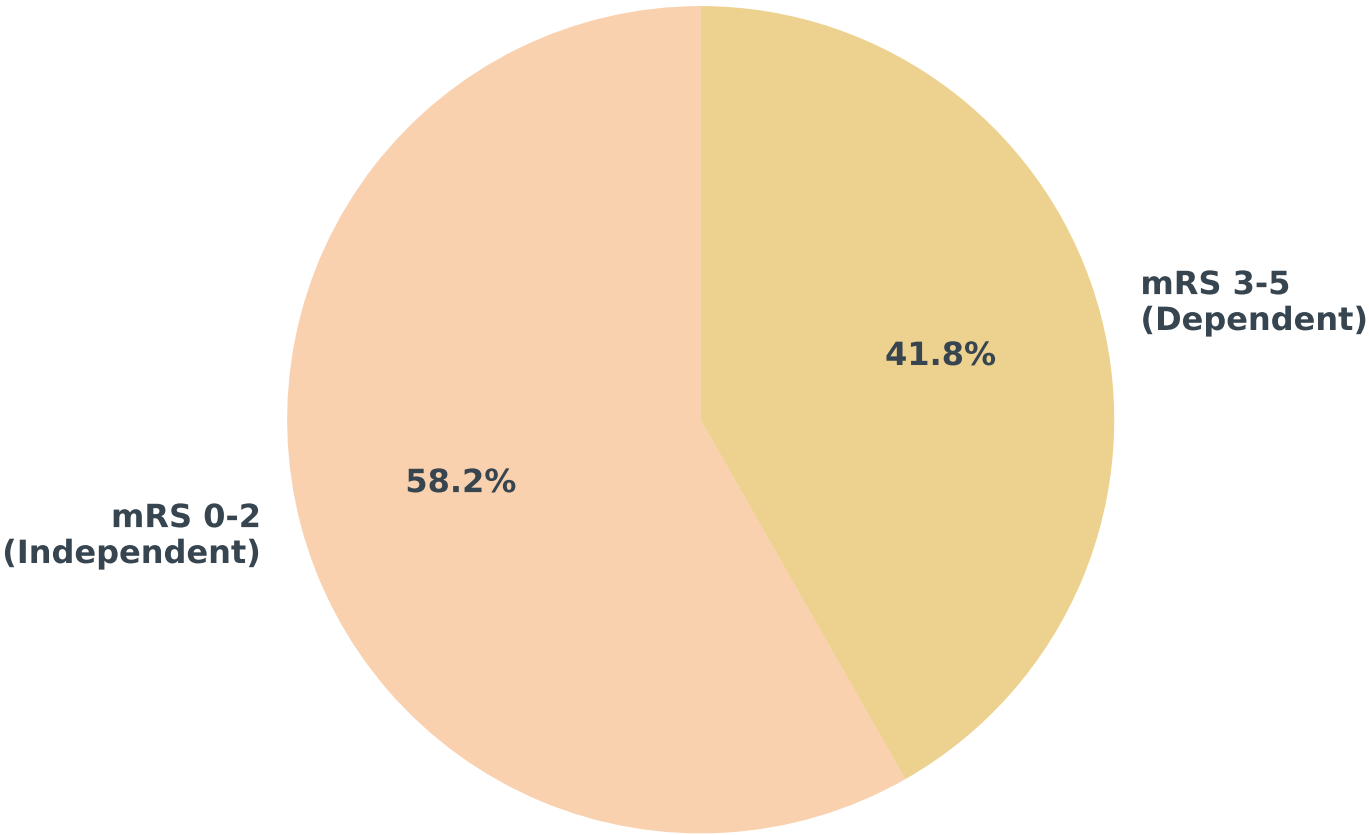
Global Impact:

- 15 million people annually experience stroke worldwide
- Leading cause of long-term disability
- Economic burden: \$721 billion annually
- 1 in 4 adults will experience stroke in their lifetime

Clinical Challenge:

- Early outcome prediction is critical
- Current methods lack accuracy and objectivity
- Need for personalized treatment strategies
- Resource allocation optimization

Need for Accurate Outcome Prediction - Detailed Analysis



Modified Rankin Scale (mRS):

- Gold standard for functional outcome assessment
- mRS 0-2: Independent (Good outcome)
- mRS 3-5: Dependent (Poor outcome)
- mRS 6: Death

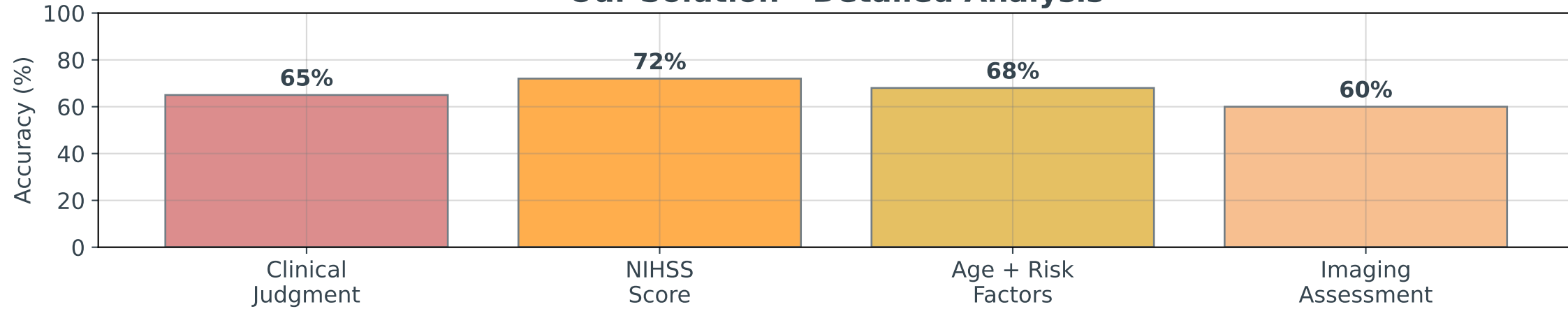
Clinical Importance:

- Guides acute treatment decisions
- Informs rehabilitation planning
- Helps family counseling
- Optimizes resource allocation
- Enables personalized care strategies

Current Limitations:

- Subjective assessment methods
- Limited early prediction accuracy
- Lack of objective biomarkers
- Inconsistent across centers

Our Solution - Detailed Analysis



Our Radiomics Approach:

- Comprehensive analysis using 455 clinical MRI scans
 - Real-world data from 91 patients (2020-2022)
 - Documented mRS outcomes for validation
 - 127 radiomic features per scan

Key Advantages:

- Objective quantification of imaging features
 - Reproducible and standardized analysis
 - Early prediction capability
 - Clinical interpretability

Expected Impact:

- Improved prediction accuracy
- Personalized treatment strategies
 - Optimized resource allocation
 - Enhanced patient counseling
 - Better rehabilitation planning